



Household Energy Dashboard

Real-time overview of your solar system performance and health

⚡ Today's Energy

28.4 kWh ▲ 12% vs yesterday



📊 Weekly Energy

164.7 kWh ▲ 8% vs last week



☀️ 28°C
Partly Cloudy

☀️ Irradiance
820 W/m²

🌬️ Wind
12 km/h

💧 Humidity
46%

🕒 Sunrise/Sunset
06:21/18:47

/ Panel Health Map (Roof Layout)



● Healthy ● Warning ● Faulty

📈 Energy loss prediction agent

Predicts future energy loss under simulated climate constraints and forecast scenarios.

Run Prediction ↗

🗣️ Real-Time Voice Agent
Interacts with your solar system through live voice interaction.

Connect to Voice Agent

📄 Root Cause Analysis Agent

analyzes the root causes of energy loss and generates panel report containing relevant panel information and actionable suggestions.

Run Analysis

Click 'Run Analysis' above to synthesize deep logical diagnosis reports.

Mastering Your AI Solar Agent

A tutorial on using your AI Solar Agent



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Interaction with AI Agent

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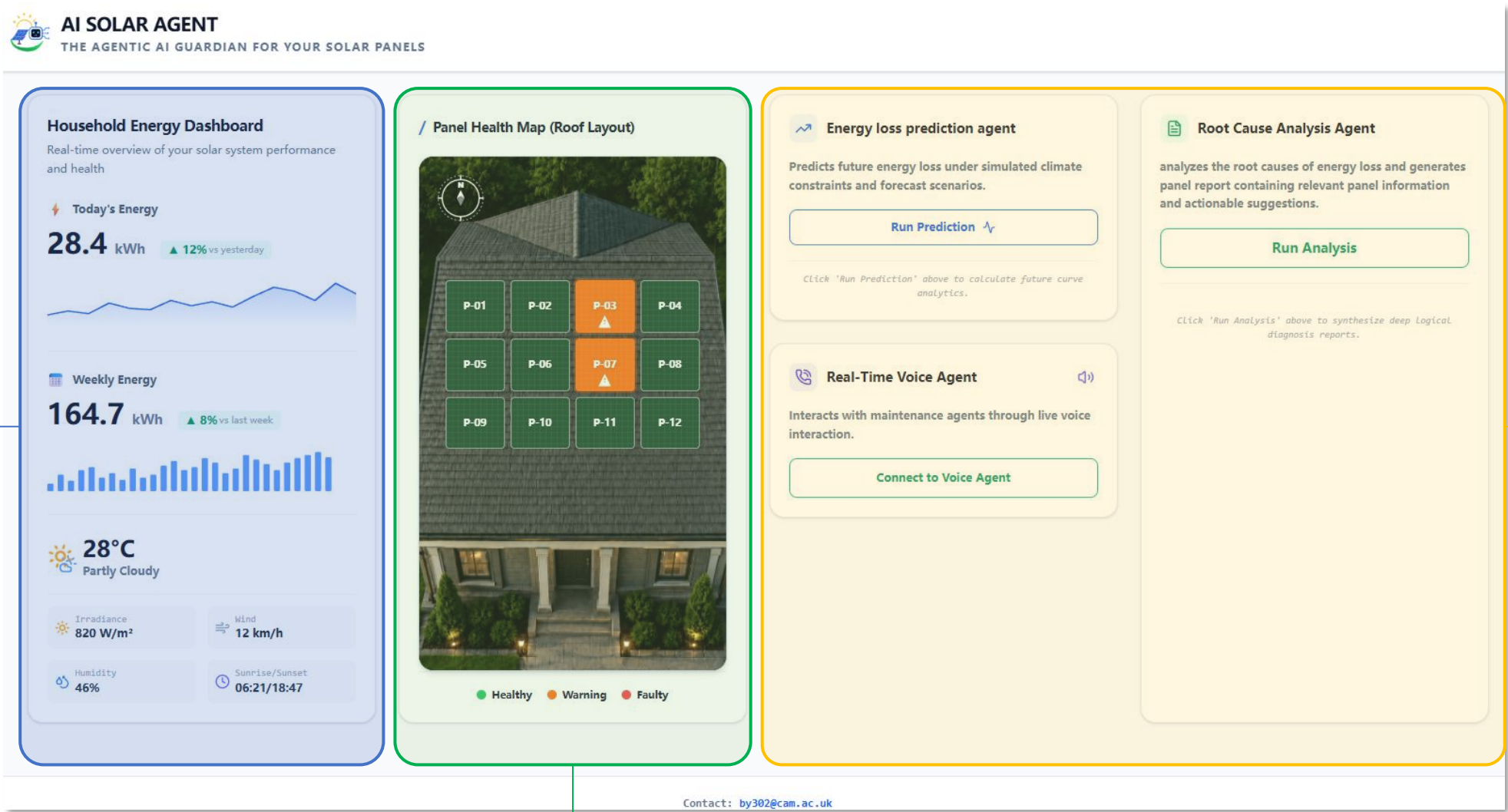
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1. Introduction to the AI Solar Agents Architecture
2. Guard your solar panels in three clicks

1. Introduction to the AI Solar Agents Architecture



Left Zone:
household
energy
dashboard

Right
Zone:
three AI
agents

Center Zone: Panel Roof Layout

2. Guard your solar panel in three clicks



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1 Step 1.1:

Click "Run Prediction" button to see the energy loss prediction results.

Contact: by302@cam.ac.uk

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ENERGY LOSS

0.8%

COMPARED WITH NORMAL OPERATIONAL ENERGY

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1 Step 1.2:

After clicking the "Run Prediction" button, you will see the energy loss prediction results. Here, 0.8% means that there is a relatively 0.8% loss compared with brand new solar panels.

Contact: by302@cam.ac.uk

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Panel Roof Layout



P-01

P-02

P-03

P-04

P-05

P-06

P-07

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P-10

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P-12

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Step 2.1:
Click "Run Analysis" button to see the root cause analysis report.

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ROOT CAUSE REPORT

Fault Diagnosis	Troubleshooting Guidance	Escalation Action
A critical observation made by the agent is the presence of brown/orange patches on the panel surface. This finding is significant because such patterns may correspond to multiple potential causes, including dirt or adhered residue, localized obstruction, persistent staining, or early signs of material degradation. Compared with obvious structural failure, this type of visual signal requires multi-hypothesis reasoning, since the exact root cause cannot be confirmed from a single front-side image. The agent therefore prioritizes contamination and localized shading as the most plausible first explanations, while still retaining internal discoloration, encapsulant aging, and other module-level degradation mechanisms as secondary possibilities.		

2

Step 2.2:
After clicking the "Run Analysis" button, you will see the root cause analysis report. Here, the report includes fault diagnosis results, troubleshooting guidance, and escalation actions.

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3 Step 3.1:

Click the "Connect to Voice Agent" button to use the voice mode to understand what is happening with your solar panels.

2. Guard your solar panel in three clicks



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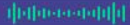
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Disconnect Agent

AGENT TRANSMITTING... LATENCY: 12MS



"hi, this is your voice agent for solar panel maintenance, according to the image of your solar panel, the root cause could be the orange patches on the panel surface, we suggest observing for another day to see if this is persistent staining and need assistance from technician."

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Step 3.2:
After clicking the "Connect to Voice Agent" button, a voice agent will start talking to you about the conditions of your solar panels, while you can see the transcript of the voice at the same time.



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Congratulations!

Now you are ready to try the AI agent by yourself at
<https://pm-psi-self.vercel.app/>